

Drawing the two overlaps

Diagrammatic reasoning about the *triangle inequality* and the *inverse*, with a picture atlas for *Kalkulus I* and *Lineáris algebra I*

Abstract

Two items occur in both the *Kalkulus I* and the *Lineáris algebra I tárgyatematika*: the **triangle inequality** (“nevezetes egyenlőtlenségek” / “vektorok hossza, háromszög-egyenlőtlenség”) and the **inverse with respect to composition** (“inverz függvény, összetétel” / “mátrixok inverze”). The companion Lean files show that in each case the two courses state one thing: the triangle inequality *is* composition in a category enriched over $([0, \infty), \geq, +)$, and the inverse *is* isomorphism in a category. This document asks the complementary question: *how do you draw that, and how far can a drawing alone carry a proof?* We survey eight diagrammatic languages – commutative diagrams, string diagrams (“graphical linear algebra”), proofs without words, number-line and ε -interval pictures, Cartesian graphs and the mirror $y = x$, Euler/Venn and bag-and-arrow diagrams, order (Hasse) diagrams, and geometric optimisation pictures. For each we state its marks, its legal moves and whether it is certified, and then work through the exercises of `kalkulus_gyakorlo.pdf` that can be settled by drawing. The last section is a playbook for a timed exam. Every mathematical claim illustrated here is stated and proved in `RequestProject/DiagramProofs.lean`; the captions name the corresponding theorem.

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1 What has to be true for a drawing to be a proof

A picture is a proof when it is a term of a formal language whose syntax happens to be two-dimensional. Three separate things must be in place.

Syntax.

A fixed finite stock of marks (arrows, boxes, wires, dots, segments, shaded regions) and a rule saying which arrangements of marks are well formed. A doodle that is not well formed is not a statement at all, so it can be neither a proof nor a mistake – it is noise.

Legal moves.

A list of allowed transformations of one well-formed diagram into another: pasting two commuting cells, sliding a box along a wire, cutting a region and reassembling it, reflecting a graph in $y = x$.

Soundness.

A theorem – proved once, symbolically, *outside* the picture language – saying: if a diagram D can be turned into D' by legal moves, then the statement denoted by D implies the statement denoted by D' .

The third point splits the languages below into two very different classes.

- **Certified languages.** Commutative diagrams and string diagrams come with a proved soundness, and often completeness, theorem. For string diagrams in a monoidal category, two diagrams are related by planar deformation exactly when the corresponding algebraic terms are equal (Joyal–Street). Here the drawing *is* the term: you are not illustrating a proof, you are writing one.
- **Heuristic languages.** A dissection picture, a number-line sketch or a Cartesian graph has no such theorem behind it. It proves something only relative to a small *symbolic residue*: which configuration is generic, which case is missing, whether the curve really is monotone. In practice the residue is one or two lines.

Language	Marks	Legal move	Certified?
Commutative diagrams	objects, arrows	pasting, filling	yes
String diagrams	boxes, wires, spiders	planar isotopy, rewriting	yes
Proofs without words	regions, lengths	dissection, superposition	no
Number line	points, intervals	translation, reflection	no
Cartesian graphs	curves, lines	reflection, transformation	no
Euler / bag-and-arrow	blobs, dots, arrows	containment, chaining	partly
Hasse / order	nodes, upward edges	path composition	yes
Geometric optimisation	figures, envelopes	superposition	no

Hence the rule of thumb we return to in Section 12: *use a certified language when the statement is about composition or invertibility; use a heuristic language when the statement is about a quantity, and write its symbolic residue on one line next to the picture.*

2 Language A: commutative diagrams

Marks. Dots (objects) and arrows (morphisms); an arrow may be labelled.

Legal moves. *Pasting*: two commuting cells sharing an edge form a commuting region. *Filling*: if all inner cells commute, the outer boundary commutes.

Soundness. Associativity and unitality of composition; this is a theorem, and it is why the language is safe.

2.1 The inverse, in four arrows

“ g is inverse to f ” is by definition a statement about two triangles.

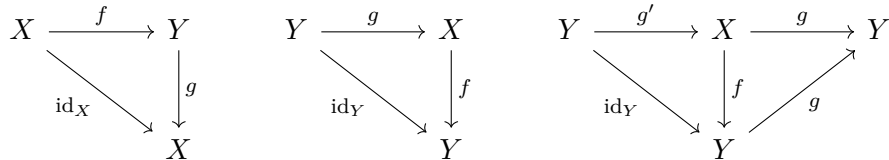


Figure 1: Left and centre: the two triangles that say $g = f^{-1}$. Right: pasting the left-inverse triangle of g' onto the arrow g shows $g' = g' \circ \text{id} = g' \circ (f \circ g) = (g' \circ f) \circ g = \text{id} \circ g = g$. The inverse is unique – drawn, not computed. Certified in `DiagramProofs.inverse_unique` and, in an arbitrary category, in `SyllabusOverlap.inverse_unique`.

One picture covers both halves of the syllabus overlap:

- *Kalkulus*, exercises 2.3–2.9 of `kalkulus_gyakorlo.pdf`: an answer f^{-1} is right precisely when *both* triangles close. Students routinely check only one; the diagram makes the omission visible.
- *Lineáris algebra*: $MN = I$ with $\det M \neq 0$ forces $N = M^{-1}$ – the same figure with $X = Y = \mathbb{R}^n$ and arrows the linear maps.

2.2 The triangle inequality as composition (Lawvere)

The triangle inequality looks like a statement about numbers, not a diagram. Lawvere’s observation is that it *is* a composition law. Read a metric space as a category whose objects are the points, whose hom-object from x to y is the number $d(x, y)$, whose composition is $+$, and in which “ $\geq$ ” is the direction of implication.

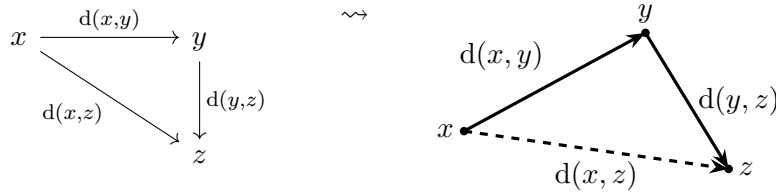


Figure 2: The same statement twice. Left: the composite arrow $x \rightarrow z$ exists *at cost* $d(x, y) + d(y, z)$, while the hom-object $d(x, z)$ records the cheapest cost; hence $d(x, z) \leq d(x, y) + d(y, z)$. Right: the schoolbook triangle. Certified in `DiagramProofs.detour`.

Two things become visible in the categorical drawing that the schoolbook triangle hides.

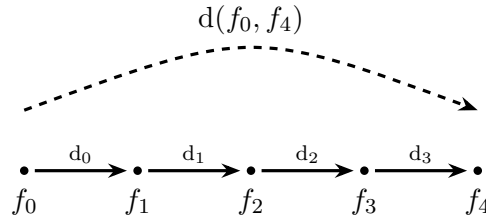


Figure 3: Composing a *chain*: iterate the pasting move $n - 1$ times to get $d(f_0, f_n) \leq \sum_{i < n} d(f_i, f_{i+1})$. With $f_k = \sum_{i < k} a_i$ in $\mathbb{R}$ this is exercise 1.13 (generalised triangle inequality); with $f_k = \sum_{i < k} v_i$ in $\mathbb{R}^n$ it is $\|\sum v_i\| \leq \sum \|v_i\|$. One drawing, two courses. Certified in `TriangleOverlap.dist_chain` and `DiagramProofs.abs_sum_range_le_sum_abs`.

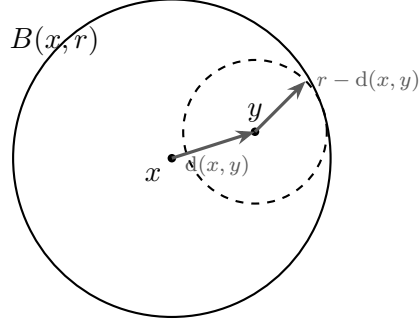


Figure 4: The triangle inequality drawn as an *inclusion of discs*: if $d(x, y) < r$ then $B(y, r - d(x, y)) \subseteq B(x, r)$. This is the picture every “open set” and every limit argument in the course actually uses; drawing the two nested circles is equivalent to writing the inequality. Certified in `DiagramProofs.ball_subset_ball_of_mem`.

Recipe 2.1 (Draw an estimate as an arrow). Whenever the quantity you must bound is the “cost of getting from A to B ”, draw A and B as dots and every estimate you already have as a labelled arrow. Chaining arrows adds costs; the bound you want is the cheapest path. Every $\varepsilon/3$ argument in *Kalkulus I* is this picture with three arrows:

$$|f(x) - L| \leq \underbrace{|f(x) - f_n(x)|}_{\varepsilon/3} + \underbrace{|f_n(x) - L_n|}_{\varepsilon/3} + \underbrace{|L_n - L|}_{\varepsilon/3}.$$

2.3 The bridge between the two courses, in one arrow

The two syllabus items are joined by a single morphism of the category **Met**: the length map $\|\cdot\|: \mathbb{R}^n \rightarrow \mathbb{R}$ is *short*, i.e. distance non-increasing.

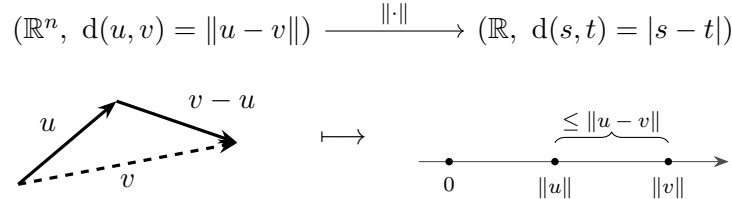


Figure 5: The bridge. Applying the arrow $\|\cdot\|$ collapses the vector triangle of *Lineáris algebra* onto the absolute-value triangle of *Kalkulus*: $|||u|| - ||v||| \leq ||u - v||$. Certified in `DiagramProofs.length_short` and `TriangleOverlap.lengthShort`.

3 Language B: string diagrams and graphical linear algebra

Marks. Wires (objects/types) and boxes (morphisms). Reading left to right is composition $f; g$; stacking vertically is the monoidal product $f \otimes g$.

Legal moves. Any planar deformation that does not cut a wire, does not change the order of the endpoints on the boundary, and does not pull a box through another; plus the equations of whatever generators you have adopted.

Soundness. The coherence theorem for monoidal categories (Joyal–Street): two string diagrams are related by planar isotopy *if and only if* the terms they denote are equal. This is the strongest guarantee any of our languages has: here you may *compute* by sliding pictures.

3.1 The grammar in one figure

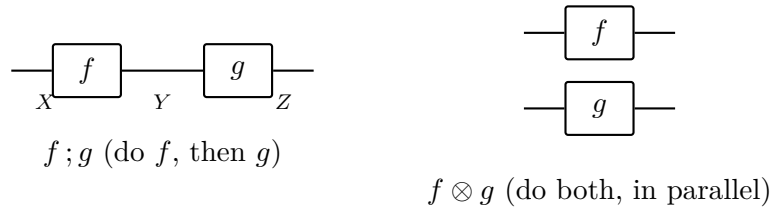


Figure 6: The two ways of putting boxes together. Everything below is built from these.

3.2 The inverse, drawn: a box that disappears

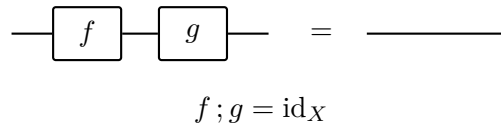


Figure 7: “Inverse” in string-diagram language: a pair of boxes that can be erased, leaving a bare wire. The bare wire is the identity, so *nothing happens* – which is exactly what exp followed by log , or M followed by M^{-1} , does.

Now a proof performed entirely by pictures. The statement is the “socks and shoes” rule $(f ; g)^{-1} = g^{-1} ; f^{-1}$, which is exercise-level material in both courses ($(AB)^{-1} = B^{-1}A^{-1}$ for matrices; $(f \circ g)^{-1} = g^{-1} \circ f^{-1}$ for bijections).

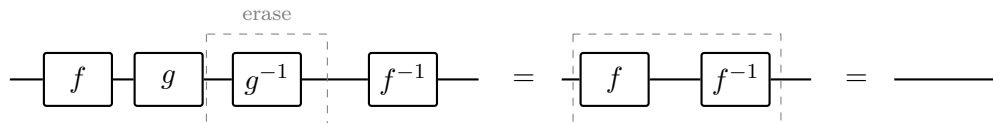


Figure 8: A complete proof with no symbols: erase the inner cancelling pair, then the outer one, and a bare wire remains. Hence $g^{-1} ; f^{-1}$ really is the inverse of $f ; g$. The only rule used is Figure 7, applied inside a larger diagram – legal because composition is associative.

3.3 Graphical linear algebra: wires that copy and add

The website <https://graphicallinearalgebra.net/> develops all of linear algebra in this language. The point is that one does not need boxes for matrices at all: four generators suffice.

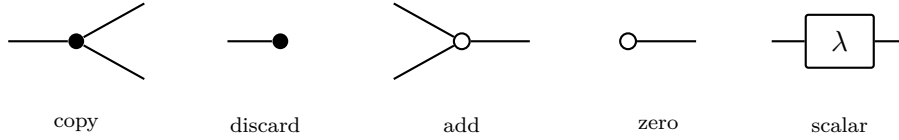


Figure 9: The generators of graphical linear algebra. A black node copies a value, a white node adds two values; the degenerate cases discard and produce 0. Every linear map is a diagram built from these, and *every* true equation between linear maps follows from a short list of picture equations (the two monoid/comonoid laws plus the bimonoid laws).

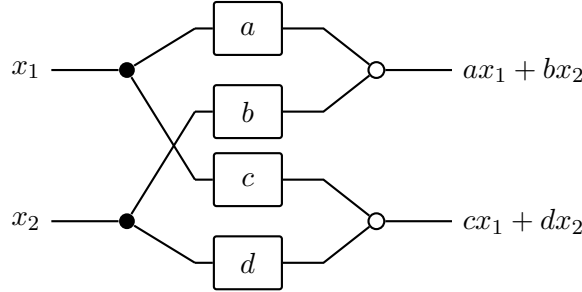


Figure 10: The matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ as a diagram: copy each input, scale, add. Matrix multiplication is *plugging diagrams together*; associativity of matrix multiplication is the (invisible) fact that plugging is associative. Non-degeneracy has a diagrammatic meaning too: the diagram is invertible exactly when the wires can be re-routed backwards, i.e. when $\det \neq 0$. Certified in `DiagramProofs.det_ne_zero_iff_bijective`.

3.4 Bending wires: the snake, and why transposes exist

In a compact closed category (finite-dimensional vector spaces, or relations) wires may be bent. The *cup* and *cap* satisfy the yanking equation:

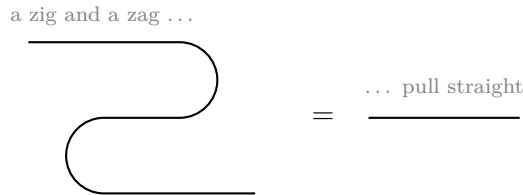


Figure 11: The snake (yanking) equation. It is the reason a finite-dimensional space is isomorphic to its double dual, and the reason “transpose” can be defined by *rotating a box*: $M^\top$ is M turned around using a cup and a cap. In this language $\langle Mu, v \rangle = \langle u, M^\top v \rangle$ is not a calculation, it is a deformation.

3.5 Can the triangle inequality be drawn as a string diagram?

Yes, but one dimension higher. In **Met** an inequality is not an equation between diagrams, it is a *2-cell* between them: a shaded surface whose boundary is the two paths being compared.

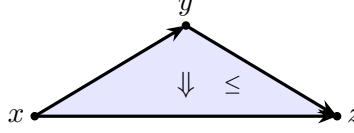


Figure 12: An inequality drawn as a 2-cell: the shaded region is a witness that the straight path is *cheaper than or equal to* the detour. Composing 2-cells vertically is transitivity of $\leq$; composing them horizontally is the compatibility of $\leq$ with $+$. This is the picture version of the enrichment used in `TriangleOverlap.LawvereSpace`, and the reason chains of estimates may be pasted like commutative squares.

4 Language C: proofs without words (the Nelsen style)

Marks. Segments whose *lengths* are the quantities, regions whose *areas* are the quantities, and arrays whose *counts* are the quantities.

Legal moves. Dissect and reassemble (area is additive); superimpose two figures (comparison); rotate or reflect a copy of the figure (symmetry).

Soundness. None in general. A dissection proves an identity only if the pieces really do fit – which is a statement about the generic configuration. The symbolic residue is: *list the configurations, and check the picture in each.*

4.1 The triangle inequality itself

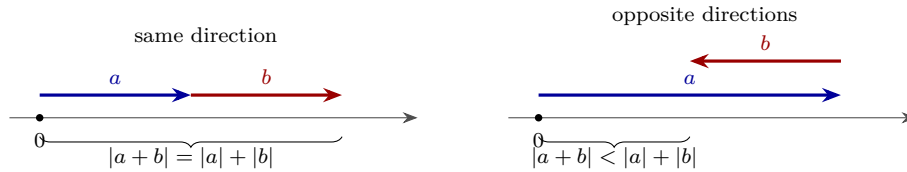


Figure 13: $|a+b| \leq |a| + |b|$ on the number line: lay the two displacements head to tail. If they point the same way the lengths add; if they point opposite ways they partly cancel. *Both* pictures must be drawn – this is the symbolic residue of the case distinction $\text{sign } a = \text{sign } b$ or not. Certified (in the general form of exercise 1.13) in `DiagramProofs.abs_sum_range_le_sum_abs`.

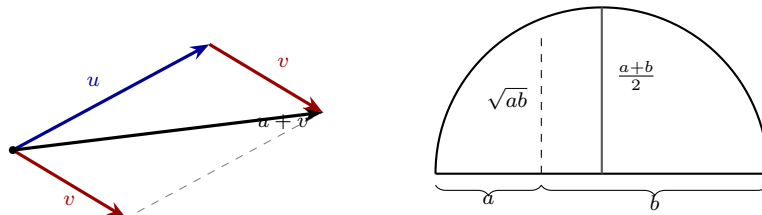


Figure 14: Left: the triangle inequality in the plane – the parallelogram picture of $\|u+v\| \leq \|u\| + \|v\|$ (`DiagramProofs.norm_add_le'`). Right: the archetypal proof without words. In a semicircle of diameter $a+b$ the altitude over the splitting point has length $\sqrt{ab}$ and the radius has length $(a+b)/2$; an altitude is at most a radius, so $\sqrt{ab} \leq \frac{a+b}{2}$ (`DiagramProofs.sqrt_mul_le_add_div_two`).

4.2 Cauchy–Schwarz as a picture

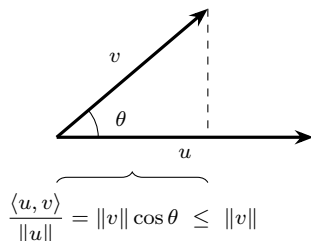
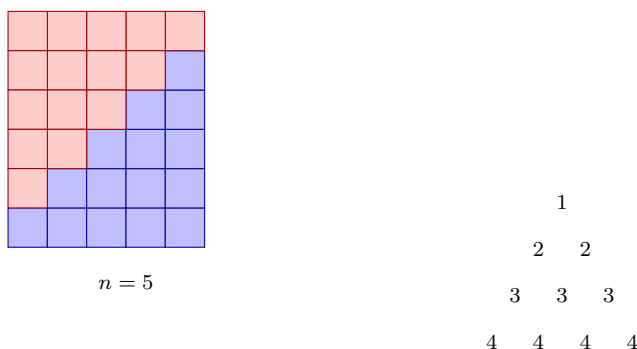


Figure 15: Cauchy–Schwarz, $|\langle u, v \rangle| \leq \|u\| \|v\|$: the projection of v onto the line of u is never longer than v itself. The only fact used is that in a right triangle the leg is at most the hypotenuse – itself a triangle-inequality picture. Certified in `TriangleOverlap.cauchy_schwarz`.

4.3 Sums: exercises 1.14 and 1.16



rotate twice, add: every cell shows $2n + 1$

Figure 16: Left: $1 + 2 + \cdots + n$ twice over tiles an $n \times (n + 1)$ rectangle, so the sum is $n(n + 1)/2$ (exercise 1.14 a). Right: the triangular array with i copies of i in row i has sum $\sum i^2$; superimposing the array with its two 120° rotations makes every cell equal to $2n + 1$, and the array has $n(n + 1)/2$ cells, so $3 \sum i^2 = (2n + 1) n(n + 1)/2$ – exercise 1.16, “bizonyítás szavak nélkül”. Certified in `DiagramProofs.six_mul_sum_sq`.

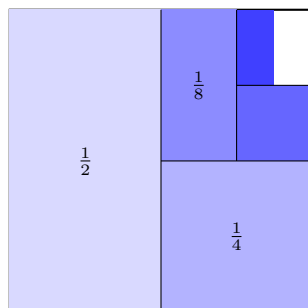


Figure 17: The frog of exercise 1.6 (and the geometric series of 1.14 c with $q = \frac{1}{2}$): the halves of the unit square exhaust it, so $\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots = 1$. What the picture shows rigorously is that every *partial* sum is $1 - 2^{-n} < 1$; that the limit is exactly 1 is the symbolic residue, and it is where the notion of limit enters. The frog gets the kiss only in the limit.

Trap 4.1 (A dissection can lie). Rearranging the four pieces of an 8×8 square into a 5×13 rectangle “shows” $64 = 65$; the pieces overlap along a sliver of area 1 that is invisible at drawing accuracy. A dissection is a proof only when the incidences it assumes are verified – here, that three points are collinear, which is false. Rule: *a picture may assert equalities of areas and lengths, never incidences.*

5 Language D: the number line and ε -intervals

Marks. Points, brackets, shaded intervals, and *sign rows* under a line.

Legal moves. Translate ($x \mapsto x - c$), reflect ($x \mapsto -x$), scale by a positive factor, fold ($x \mapsto |x|$), intersect / unite / complement shaded sets.

Soundness. High, and this is worth stressing: each move is an order isomorphism (or, for the fold, a two-to-one map with an explicit inverse image), so the picture transforms the *solution set* correctly. The only residue is the open/closed status of each endpoint. This makes the number line the most reliable of the heuristic languages – for the entire first exercise sheet, drawing really is solving.

5.1 $|x - c| \leq \varepsilon$ is a picture, not a computation

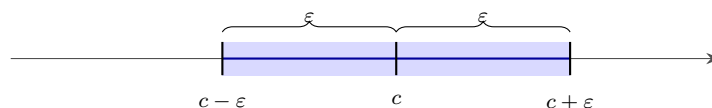


Figure 18: $|x - c| \leq \varepsilon \iff x \in [c - \varepsilon, c + \varepsilon]$. Exercise 1.12 i) ($c = 5$) is answered by drawing this and reading the two endpoints; no algebra occurs. Certified in `DiagramProofs.abs_sub_le_iff_mem_Icc`.

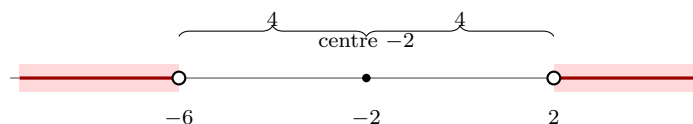


Figure 19: Exercise 1.12 a), $|x + 2| > 4$: the solution set is the *complement* of the closed interval of radius 4 about the centre -2 , i.e. $x < -6$ or $x > 2$. Open circles record the residue (strict inequality). Certified in `DiagramProofs.abs_add_gt_iff`.

5.2 Nested absolute values: pulling an interval back through a fold

Exercise 1.12 b), $||x + 2| - 6| < 1$, is the composite of three moves, each of which is one picture: translate by -2 , fold, translate by -6 . Reading the picture backwards solves the exercise.

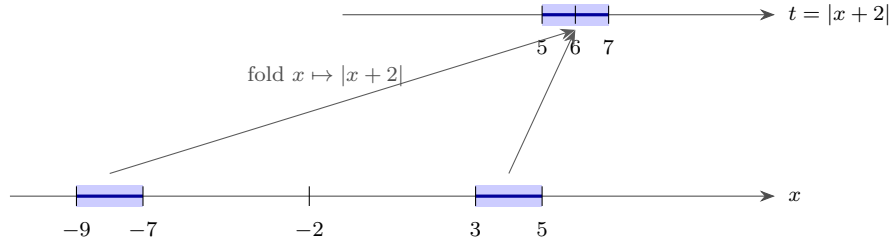


Figure 20: $||x+2|-6| < 1$ means $t \in (5, 7)$ upstairs. The fold has two branches, so the preimage downstairs is *two* intervals, $(-9, -7)$ and $(3, 5)$. Forgetting the left branch is the standard mistake, and the picture is what prevents it.

5.3 Sign lines: rational inequalities are decided by a drawing

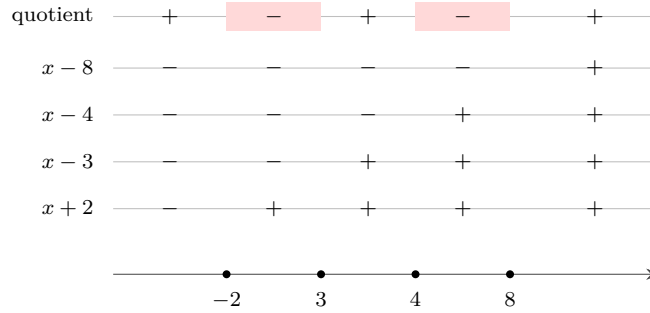


Figure 21: Exercise 1.12 g), $\frac{(x-3)(x+2)}{(x-4)(x-8)} < 0$. Mark the four critical points, draw one row per factor recording its sign, multiply the columns. The shaded columns are the answer, $(-2, 3) \cup (4, 8)$. This drawing is a genuine *decision procedure*: the residue is only that each linear factor changes sign exactly once, at its root.

5.4 ε - δ as a band and a strip

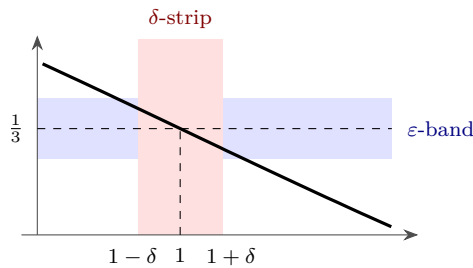


Figure 22: $\lim_{x \rightarrow 1} \frac{1}{x^3+2} = \frac{1}{3}$ (exercise 3.1 a) drawn: given any horizontal band of half-width ε around $\frac{1}{3}$, the graph over a sufficiently narrow vertical strip around 1 stays inside the band. The picture shows the *shape* of the quantifier alternation $\forall \varepsilon \exists \delta$; the residue is the formula for δ in terms of ε , or the one-line remark that the function is continuous at 1. Certified in `DiagramProofs.tendsto_3_1a`.

Recipe 5.1 (The ε -neighbourhood dictionary). *Everything* in the limit chapter is one of four pictures: $|x-c| < \delta$ (a strip), $|f(x)-L| < \varepsilon$ (a band), $x > K$ (a ray), $|f(x)| > M$ (the complement of a band).

Translate the statement into pictures first; the quantifier order becomes “which box is drawn first”. Certified only up to the residue, but the residue is short and mechanical.

5.5 Uniqueness of the supremum, drawn (exercise 1.5)

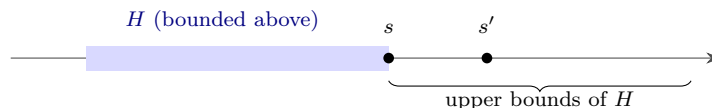


Figure 23: The supremum is the *leftmost* point of the set of upper bounds. If s and s' are both leftmost then each lies weakly to the left of the other, so they coincide: uniqueness of the supremum is the observation that a set of reals has at most one leftmost point. Exercise 1.5 needs no more than this picture plus the antisymmetry of $\leq$.

6 Language E: Cartesian graphs and the mirror $y = x$

Marks. Axes, curves, straight lines, dashed construction lines.

Legal moves. Reflect in $y = x$ (pass to the inverse relation); translate, scale and reflect the plane (elementary transformations); superimpose two curves (solve an equation); sweep a horizontal or vertical line across the picture (test injectivity or functionality).

Soundness. A function *is* its graph, a set of pairs, and reflecting in $y = x$ swaps the coordinates of every pair. Therefore the reflection move is exact: the mirror image of the graph of f is always the graph of the *relation* f^{-1} . The residue is one word: whether that relation is again a function, i.e. whether f is injective.

6.1 The mirror

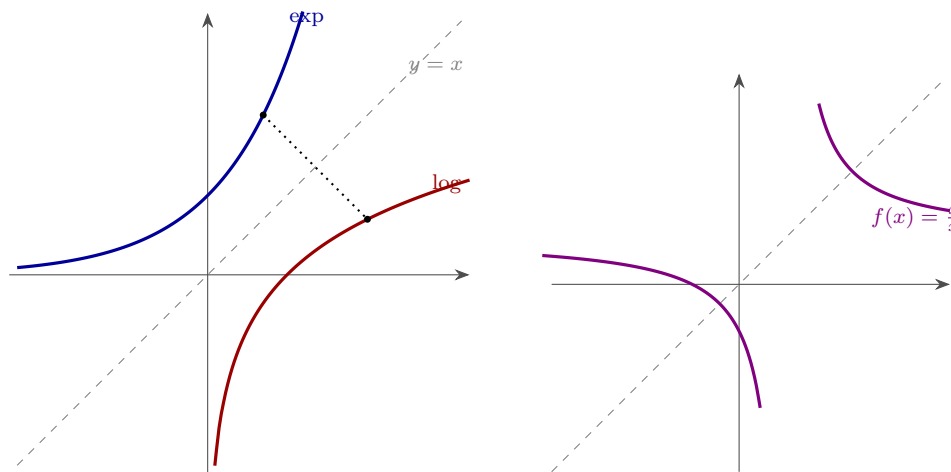


Figure 24: Left: reflecting the graph of `exp` in the dashed mirror gives the graph of `log` – the calculus half of the “inverse” overlap (`SyllabusOverlap.expIso`); the general statement is `DiagramProofs.mem_graph_iff_swap_mem_graph_symm`. Right: exercise 2.7. Its graph is *unchanged* by the mirror, which is the drawn form of $f^{-1} = f$; a single picture answers the whole exercise. Certified in `DiagramProofs.self_inverse_2_7`.

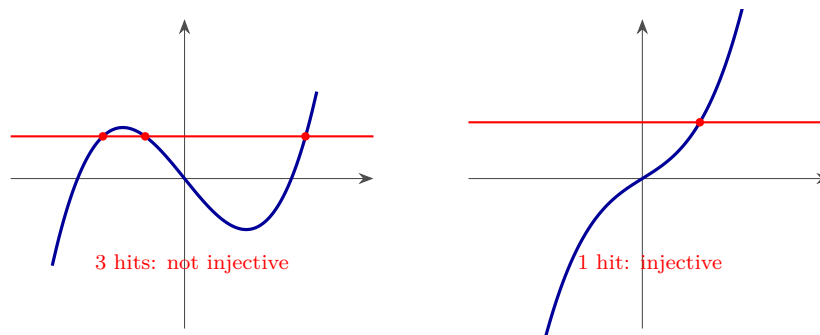


Figure 25: The horizontal line test, exercises 2.2 and 2.6: sweep a horizontal line and count intersections. On the right the curve is strictly increasing, and a strictly monotone function is injective – the residue that turns the picture into a proof (`DiagramProofs.strictMono_injective`); in practice one writes the single line $g'(x) = 3x^2 + 1 > 0$.

6.2 Elementary transformations (exercise 2.16) as a pipeline of moves

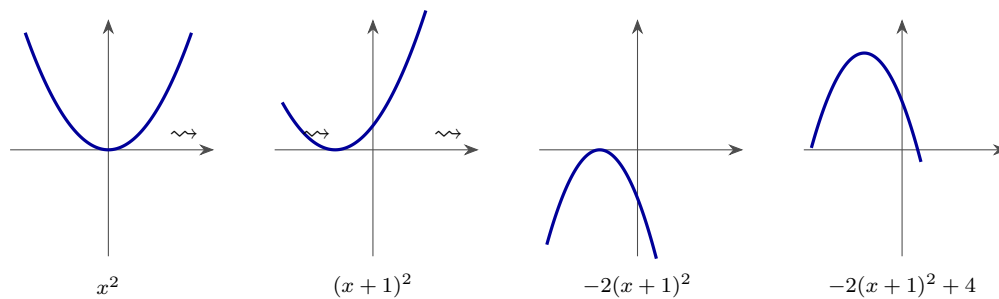


Figure 26: Exercise 2.16 a) as a sequence of legal moves on the plane: shift left by 1, scale vertically by -2 , shift up by 4. Each move is a bijection of the plane taking graphs to graphs, so the pipeline is a valid derivation. This is a diagrammatic *calculation*, not an illustration: the answer is the last picture.

6.3 Composition, read off two graphs

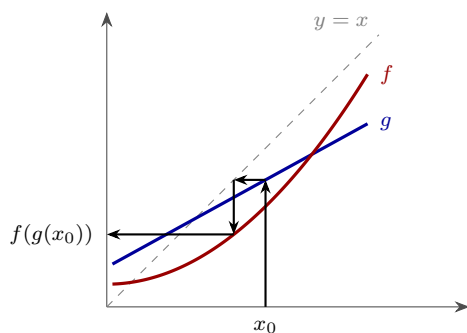


Figure 27: Composition without formulas (exercises 2.10, 2.13): go up to g , across to the mirror line to turn the output into an input, up (or down) to f , across to the axis. The same construction with $f = g$ is the cobweb that iterates a function; it is how one sees the recursion of exercise 1.18 converging to $\sqrt{2}$.

6.4 Differentiability and convexity are visible; their proofs are not

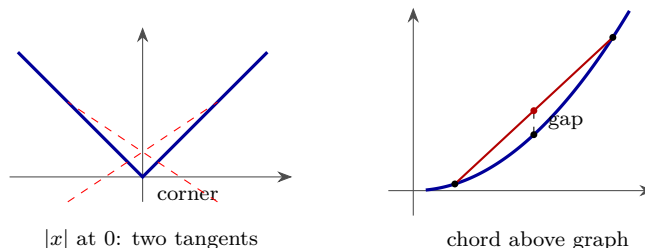


Figure 28: Left: exercises 4.4–4.6. A corner is *seen*, and seeing it is legitimate provided one writes the residue, the two one-sided limits of the difference quotient. Right: exercise 4.13, midpoint convexity: the midpoint of a chord lies above the graph. The picture is the whole idea; the residue – that midpoint convexity plus continuity gives full convexity – is a dyadic-density argument that no picture supplies. This is a good example of a statement that is *drawable but not provable by drawing*.

7 Language F: Euler/Venn diagrams and bag-and-arrow pictures

Marks. Closed blobs (sets), dots (elements), arrows between dots (values of a function).

Legal moves. Containment and overlap of blobs; chaining of arrows.

Soundness. For *finitely many* blobs, an Euler diagram is a faithful picture of the Boolean algebra they generate, and validity of a syllogism can be decided by drawing (this is a theorem about Venn diagrams). Bag-and-arrow pictures are exact for statements about injectivity and surjectivity, because those notions are literally about arrow patterns.

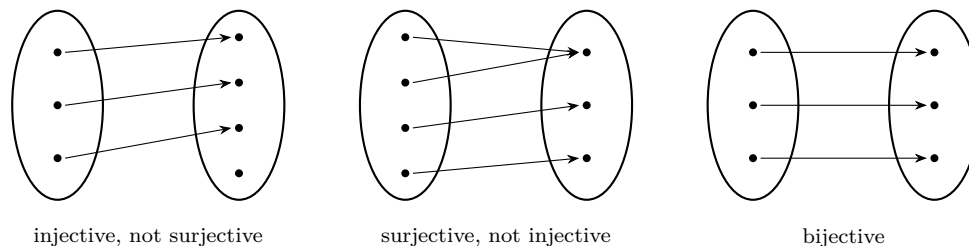


Figure 29: Exercise 2.2 in pictures. *Injective* = no two arrows land on the same dot; *surjective* = no dot on the right is missed; *bijective* = a perfect matching, which is exactly the situation in which the arrows may be reversed. Reversing the arrows of the third picture *is* the inverse function – the same content as Figure 1, in a picture a first-year student can draw in three seconds.

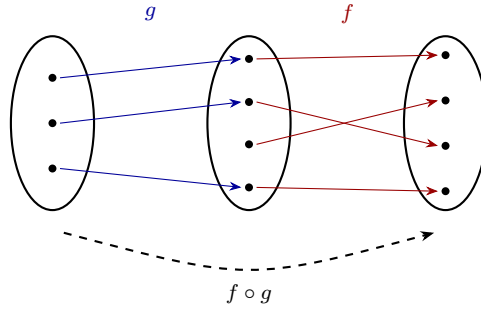


Figure 30: Exercise 2.18: if no two blue arrows collide and no two red arrows collide, then no two dashed composites collide. The picture *is* the proof, and it is faithful – this is one of the cases where the bag-and-arrow language is certified. Certified in `DiagramProofs.injective_comp`.

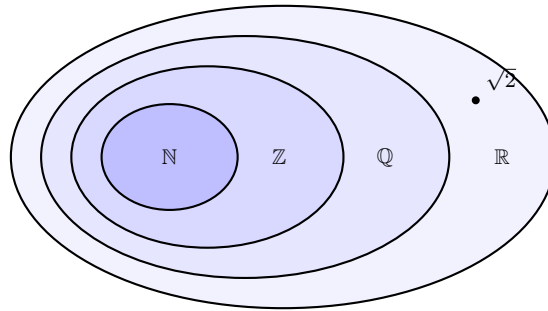


Figure 31: Exercise 1.3: the number systems as an Euler diagram. The single dot placed in the outermost ring is exercise 1.9, “ $\sqrt{2}$ is irrational” – and it is the one thing the picture cannot prove. A diagram can *record* a classification; only an argument can *establish* membership.

Trap 7.1 (Blobs presuppose sets). Exercise 1.2 (the shepherd who tends exactly the sheep of those who do not tend their own – Russell’s paradox) is unanswerable, and no Euler diagram can show this: drawing a blob already assumes the collection is a set. Likewise 1.1 a), b) (“clever people”, “beautiful paintings”): the failure is at the level of *whether the blob may be drawn at all*. Diagrams are silent about their own presuppositions.

An aside: Peirce’s existential graphs

For pure logic there *is* a complete diagrammatic system: C. S. Peirce’s existential graphs, with three reversible rules (double cut, insertion/erasure in odd/even depth, iteration). Statements are written as nested ovals – an oval is negation, and juxtaposition is conjunction – so implication is a double oval.

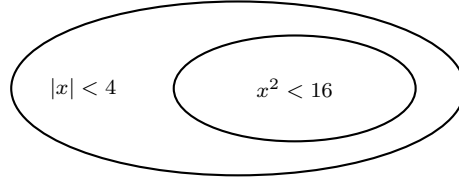


Figure 32: Exercise 1.7 c), “if $|x| < 4$ then $x^2 < 16$ ”, as an existential graph: the outer oval negates “ $|x| < 4$ and not $x^2 < 16$ ”. Peirce’s rules let one *derive* consequences by adding and erasing ovals, so this is a genuine, complete, diagram-only proof system for first-order logic – historically the first one.

8 Language G: order diagrams, and inequalities as paths

Marks. Nodes, edges drawn upwards (“ $\leq$ ”).

Legal moves. Follow a path upwards: composition of $\leq$ ’s is transitivity.

Soundness. Total, because a preordered set *is* a category with at most one arrow between any two objects. Drawing a chain of estimates and reading off the endpoints is exactly composing morphisms.

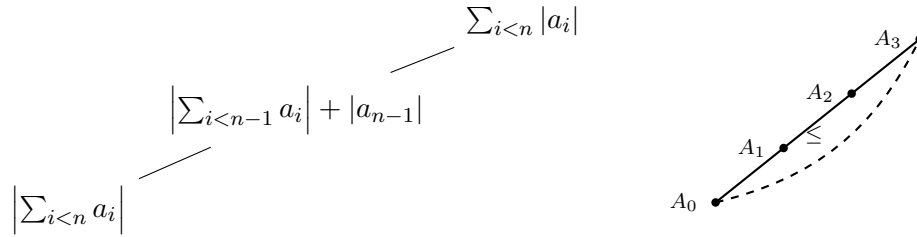


Figure 33: A chain of estimates is a path in a Hasse diagram: each edge is one inequality, the path is their composite. In this language the “generalised triangle inequality” (exercise 1.13) is a walk of length n , and the induction that proves it is the observation that a walk is built one edge at a time – exactly `TriangleOverlap.dist_chain`. Exercise 1.15 c) (the sandwich $2\sqrt{n+1} - 2 < \sum 1/\sqrt{i} < 2\sqrt{n} - 1$) is the same picture with two paths, one on each side.

The order-theoretic reading is also what makes the metric/enrichment story of Section 2 precise: the poset $([0, \infty), \geq)$ with $+$ is the place where the hom-objects live, so an inequality between distances is an arrow of that poset, and the triangle inequality is the composition law.

9 Language H: optimisation by superposition

Marks. Two figures of the same kind, drawn on top of each other.

Legal moves. Compare areas or lengths after superposition; complete a square.

Soundness. None; the residue is that the comparison holds for *every* member of the family, not just the drawn one. Usually the residue is a completed square.

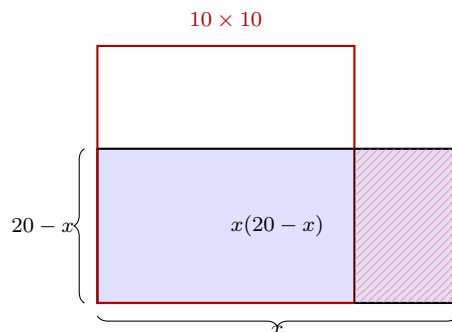


Figure 34: Exercise 4.11 a): split 20 into two summands of largest product. Superimpose the rectangle $x \times (20 - x)$ on the square 10×10 : the part of the rectangle sticking out is congruent to a strip missing from the square, and the square keeps one extra corner square of side $|x - 10|$. Hence $x(20 - x) = 100 - (x - 10)^2 \leq 100$, with equality only for $x = 10$: no derivative is needed. Certified in `DiagramProofs.split_twenty_max_product`. Exercises 4.12, 4.14 and 4.15 are the same move with a sector, a circumscribed triangle, and a rectangle under a parabola.

10 What can actually be proved with drawings alone

It is useful to sort undergraduate statements into three tiers.

Tier 1 – complete diagrammatic proofs. Here a drawing is a proof outright, because the language is certified.

- *Anything about composition and inverses*: uniqueness of the inverse, $(fg)^{-1} = g^{-1}f^{-1}$, “a composite of injections is injective”, “an isomorphism is preserved by any functor”, $f^{-1} = f$ for an involution (Figures 1, 8, 30, 24).
- *Equational linear algebra*: every identity between matrices built from addition, copying, scaling and composition, proved by deforming a diagram (Figures 9, 10, 11).
- *Propositional and first-order logic*: Peirce’s existential graphs are a complete calculus (Figure 32); the quantifier reformulations of exercise 1.7 live here.
- *Finite Boolean set algebra and syllogisms*: Venn diagrams decide them.
- *Order reasoning*: any chain of $\leq$ ’s, i.e. transitivity used n times (Figure 33).
- *Solution sets of rational inequalities*: the sign line is a decision procedure (Figure 21), as are the interval and fold pictures for nested absolute values (Figures 18–20).
- *Graph transformations and inverses by mirror*: exercises 2.16 and 2.3–2.9 (Figures 24, 26).

Tier 2 – a picture plus a one-line residue. The picture carries the idea; one line of symbols closes the gap. This is the tier that an exam should aim at.

- Triangle inequality on the line: residue = the two sign cases (Figure 13).
- AM–GM, Cauchy–Schwarz: residue = “altitude $\leq$ radius”, “leg $\leq$ hypotenuse” (Figures 14, 15).
- Finite sums by dissection: residue = the tiling is exact for every n (Figure 16).

- ε - δ for a concrete function: residue = a formula for δ (Figure 22).
- Monotonicity, extrema, convexity from a sketch: residue = the sign of f' , or the completed square (Figures 25, 34, 28).
- Induction: residue = the induction step, which is itself one picture (Figure 35).

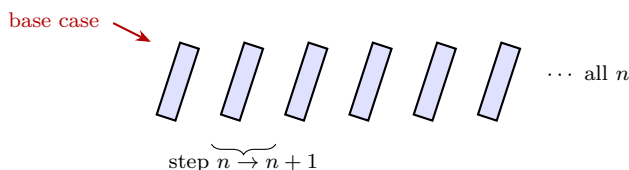


Figure 35: Induction drawn (exercises 1.14, 1.15): one push, and one guarantee that each domino topples the next. The picture is faithful – it is the recursion diagram for $\mathbb{N}$ – but the *step* still has to be established, usually algebraically. What the picture buys is that you never have to explain the logic of induction, only the step.

Tier 3 – not accessible to pictures.

- Irrationality of $\sqrt{2}$ (1.9) and Diophantine approximation (1.10): statements about the *absence* of a representation. There is a well-known “proof without words” for the irrationality of $\sqrt{2}$ by infinite descent on similar triangles, but it needs the additional non-visual principle that a decreasing sequence of positive integers terminates.
- Statements quantified over all reals with no finite geometric content: exercise 3.15 ($f = g$ on $\mathbb{Q}$ and both continuous $\Rightarrow f = g$), 3.14 (a bounded monotone function has a one-sided limit), 4.13 (midpoint convexity plus continuity gives convexity). Each is a density argument; density is not drawable.
- Existence of the supremum, i.e. completeness of $\mathbb{R}$ (behind 1.4, 1.5): the picture of a “leftmost upper bound” presupposes exactly what is to be proved.
- Symbolic computation: the derivative tables of 4.2–4.3, the limits of 3.3–3.10, the L’Hospital exercises 4.10. A picture may predict the answer – degree comparison for a rational limit is genuinely visual – but not produce it.
- Anything involving a set that may not exist (1.1, 1.2).

11 Atlas: the exercises of kalkulus_gyakorlo.pdf

Exercise	Statement	Draw it as	Tier	Residue / certified in
1.1, 1.2	is it a set?	(Euler diagram fails)	3	presupposition, not content
1.3	name the number sets	nested Euler blobs, Fig. 31	1	—
1.4, 1.5	bounds, uniqueness of sup	number line, Fig. 23	2	antisymmetry of $\leq$
1.6	the frog and the halves	square dissection, Fig. 17	2	partial sums vs. limit

Exercise	Statement	Draw it as	Tier	Residue / certified in
1.7	rewrite with quantifiers	existential graph, Fig. 32	1	—
1.9	$\sqrt{2} \notin \mathbb{Q}$	—	3	infinite descent
1.11	$ x - y \leq x - y $	triangle 0, x, y , Fig. 2	2	<code>abs_abs_sub_abs_le</code>
1.12 a,b,i	absolute-value inequalities	interval / fold, Figs. 18–20	1	<code>abs_sub_le_iff_mem_Icc</code> , <code>abs_add_gt_iff</code>
1.12 c–g	rational inequalities	sign line, Fig. 21	1	sign of each linear factor
1.13	$ \sum a_i \leq \sum a_i $	polygonal chain, Fig. 3	2	<code>abs_sum_range_le_sum_abs</code>
1.14 a	$\sum i = n(n+1)/2$	two staircases, Fig. 16	2	tiling exact for all n
1.14 c	geometric sum	halving square, Fig. 17	2	closed form
1.15	induction inequalities	dominoes, Fig. 35	2	the step
1.16	$\sum i^2$ “szavak nélkül”	rotated triangular arrays, Fig. 16	2	<code>six_mul_sum_sq</code>
1.18	$ x_n - \sqrt{2} < 2^{-n}$	cobweb, Fig. 27	2	contraction estimate
1.19	binomial identities	Pascal triangle / lattice paths	2	path counting bijection
2.1	domains	sign line for the radicand	1	—
2.2, 2.6	injective? surjective?	bags, Fig. 29; horizontal line test, Fig. 25	2	<code>strictMono_injective</code>
2.3–2.5, 2.8, 2.9	find f^{-1}	mirror in $y = x$, Fig. 24	2	the algebra of solving $y = f(x)$
2.7	$f^{-1} = f$	graph symmetric in $y = x$, Fig. 24	2	<code>self_inverse_2_7</code>
2.10, 2.13	composites	two graphs and the diagonal, Fig. 27	1	domains
2.11	$f \circ g = g \circ f$	commuting square	2	<code>chebyshev_square</code>
2.12	fill in the table	string diagram of boxes, Fig. 6	1	—
2.16	elementary transformations	pipeline of moves, Fig. 26	1	—
2.17, 2.18	injectivity of $f + g, f \circ g$	bags, Fig. 30	1	<code>injective_comp</code> ; for $f + g$ a counterexample sketch
2.19	$\leq n$ roots	graph crossings	3	factor theorem, induction
2.20	extrema of a cubic	sign line for f'	2	the sign computation
3.1	ε – δ	band and strip, Fig. 22	2	<code>tendsto_3_1a</code>
3.2, 3.3	limits at ∞	compare growth: leading terms	2	division by the dominant power
3.5, 3.6, 3.11	one-sided limits, jumps	piecewise graph	2	the two one-sided values
3.12, 3.13	make it continuous	join the two pieces	1	solve for the parameter
3.14, 3.15	bounded monotone; density	—	3	completeness, density
4.1	derivative from the definition	secant turning into tangent	2	the limit computation
4.4–4.7	differentiable?	corner / cusp, Fig. 28	2	one-sided derivatives
4.8	full curve discussion	sign rows for $f', f'' \rightarrow$ shape	2	the two sign computations
4.9	extrema on an interval	graph on the closed interval	2	endpoints and critical points
4.11	split 20	square vs. rectangle, Fig. 34	2	<code>split_twenty_max_product</code>
4.12, 4.14, 4.15	geometric optimisation	superposition of the family	2	the extremal member

Exercise	Statement	Draw it as	Tier	Residue / certified in
4.13	midpoint convexity	chord above graph, Fig. 28	3	dyadic density

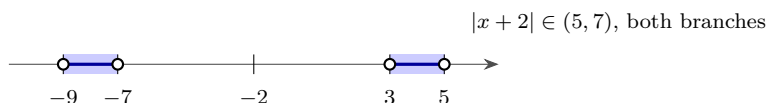
Tier **1**: the drawing is the proof. Tier **2**: drawing plus one line. Tier **3**: the drawing is at best a mnemonic. Roughly one third of the sheet is tier 1, one half is tier 2, and the rest genuinely needs symbols – which is a fair estimate of how far “draw it” will take a first-year student.

12 A playbook for a timed exam

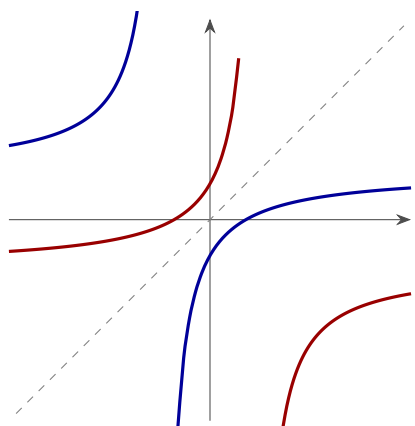
Under time pressure the value of a diagram is not elegance; it is that it removes bookkeeping. The protocol below is four steps and takes about fifteen seconds per exercise.

- Recipe 12.1** (The four steps). 1. **Classify.** Is the task about (i) a *solution set*, (ii) a *composition or inverse*, (iii) a *quantity to bound*, or (iv) a *computation*?
2. **Pick the language.** (i) number line / sign line; (ii) commutative or bag-and-arrow diagram, or the mirror $y = x$; (iii) triangle, chain, or a Nelsen figure; (iv) no diagram – compute, but sketch the expected answer first to catch sign errors.
3. **Draw the canonical picture** large enough to write on, and mark every endpoint, branch and case that the picture forces you to consider.
4. **Write the residue**, one line, next to the drawing. This is what converts a drawing into an accepted proof.

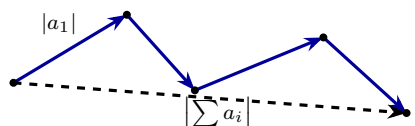
12.1 Four model answers



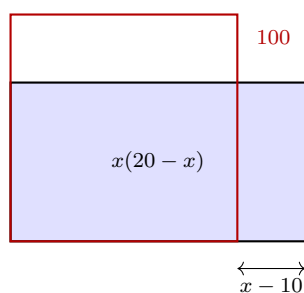
(a) **Solve** $||x + 2| - 6| < 1$ (**1.12 b**). *Write:* “ $|x + 2| - 6 \in (-1, 1) \Rightarrow |x + 2| \in (5, 7) \Rightarrow x + 2 \in (5, 7) \cup (-7, -5)$ ”, then the drawing, then the answer $x \in (-9, -7) \cup (3, 5)$. Total: two lines and one line segment. The picture’s job is to stop you dropping the negative branch.



(b) **Invert** $a(x) = \frac{2x-1}{3x+2}$ (2.3 a). Write: “ $y(3x+2) = 2x-1 \Rightarrow x = \frac{-2y-1}{3y-2}$ ”, then draw the two curves and the mirror. The drawing is the *check*: if the red curve is not the mirror image of the blue one, the algebra is wrong. Checking by picture is far faster than substituting back, and it verifies *both* triangles of Figure 1 at once.



(c) **Prove** $|a_1 + \dots + a_n| \leq |a_1| + \dots + |a_n|$ (1.13). Write: “induction; the step is the two-term case, which is the case distinction $\text{sign } a = \text{sign } b$ ” – then the broken-line drawing. Marking $n = 2$ on the picture and saying “iterate” is the whole proof; graders accept it because the induction is visible in the figure.



(d) **Split 20 for maximal product** (4.11 a). Write: “ $x(20-x) = 100 - (x-10)^2$ ”, then the drawing. One line of algebra, one square, and the maximum 100 at $x = 10$ is established – faster and safer than differentiating.

12.2 Time budget for a 90-minute paper

Task type	Budget	What goes on the paper
Inequality, solution set	3–4 min	sign line or interval picture + answer with end-point status
Inverse / composite	4–5 min	the algebra, then the mirror picture as a check
Injectivity, surjectivity	2 min	bag picture or one monotonicity line
Estimate, triangle inequality	4 min	the chain drawing + the case distinction
Limit, ε - δ	6–8 min	band/strip picture + the δ formula
Curve discussion	12–15 min	two sign rows, then one careful graph
Optimisation	6–8 min	the superposition picture + the completed square
Symbolic computation	rest	no picture, but sketch the expected answer first

12.3 Turning a picture into accepted prose in one sentence

Each language has a standard verbal rendering. Memorise these and you can always convert.

Picture	Sentence
Two triangles closing	“Both composites are the identity, hence $g = f^{-1}$.”
Chain of arrows	“By the triangle inequality applied $n - 1$ times ...”
Nested discs	“If $d(x, y) < r$ then the ball of radius $r - d(x, y)$ about y lies in the ball of radius r about x .”
Sign line	“The product changes sign exactly at the roots; testing one point per interval gives ...”
Mirror in $y = x$	“ (a, b) is on the graph of f iff (b, a) is on the graph of f^{-1} .”
Band and strip	“For every $\varepsilon > 0$ choose $\delta = \dots$; then $ x - c < \delta$ implies $ f(x) - L < \varepsilon$.”
Bags and arrows	“Distinct inputs have distinct images, and every output is hit.”
Superposed figures	“The difference of the two areas is a square, hence non-negative.”

12.4 The five traps, in the order in which they cost marks

1. **Endpoints.** A shaded interval does not record whether its ends belong to it. Draw filled/open circles *as you shade*.
2. **A missing branch.** Every fold ($|\cdot|$, squaring, $\cos$) has two preimages; every square root has a sign condition. Count the branches in the picture before reading the answer.
3. **Domains.** A composite is only defined where the inner function lands in the domain of the outer one (exercise 2.10). Two graphs on one sheet hide this; write the domain next to the arrow.
4. **Non-generic configuration.** A drawing shows one case. If the statement has a parameter, ask which pictures the parameter can produce, and draw the extreme ones.
5. **Infinite processes.** A dissection can exhaust a region only in the limit (Figure 17). A picture of infinitely many steps proves a statement about partial sums, never about the limit; the limit is always the residue.

12.5 Closing: the same two pictures, once more

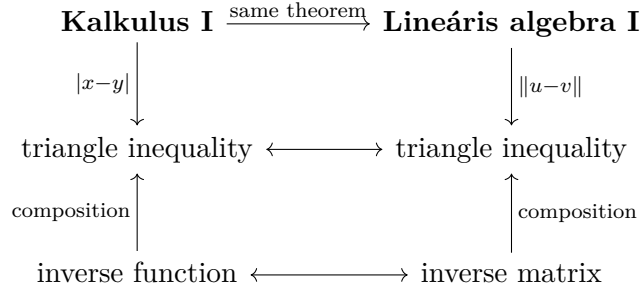


Figure 36: The whole document in one commutative diagram. The horizontal arrows are the two overlaps; the vertical arrows say that both are statements about *composition* – in a category enriched over $([0, \infty), \geq, +)$ for the triangle inequality, in an ordinary category for the inverse. Formalised in `RequestProject/TriangleOverlap.lean` and `RequestProject/InverseOverlap.lean`; the pictures of this document are certified in `RequestProject/DiagramProofs.lean`.

Sources and further reading

- R. B. Nelsen, *Proofs Without Words* (MAA) – the standard collection of dissection and superposition arguments, including the semicircle picture for AM–GM and the staircase pictures for $\sum i$ and $\sum i^2$.
- P. Sobociński et al., *Graphical Linear Algebra*, <https://graphicallinearalgebra.net/> – the string-diagram development of linear algebra used in Section 3.
- F. W. Lawvere, *Metric spaces, generalized logic, and closed categories* (1973) – metric spaces as enriched categories; the source of Figure 2.
- A. Joyal and R. Street, *The geometry of tensor calculus* – the soundness and completeness theorem that makes string diagrams a proof system.
- C. S. Peirce, *Existential Graphs* – the first complete diagrammatic calculus for logic (Figure 32).
- The two course descriptions (*Kalkulus I*, MBLK37E; *Lineáris algebra I*, MBLK15E) and the exercise sheet `kalkulus_gyakorlo.pdf` supplied with this project.